Bence Csiba

London, UK | linkedin.com/in/bencecsiba | 07496063113 bencecsiba5@gmail.com

WORK EXPERIENCE

AstraZeneca

AI Engineer *2024 - Present*

- Part of the research-focused Deep Learning Team within R&D IT, contributing to a pod focused on the volumetric segmentation of lesions on 3D medical imaging, primarily CT scans, as well as PET scans.
- Worked with radiologists and front-end engineers to deploy models on an internal radiology platform to optimise the radiologists' experience and model performance.
- Co-authored LinGuinE, a package to speed up the segmentation by radiologists of tumours by propagating guidance clicks through longitudinal studies. (https://github.com/n-e-garibli/LinGuinE)
- LinGuinE received early acceptance to MICCAI 2026 (27.5% acceptance rate) and shortlisted for best paper (top 25 papers) (https://papers.miccai.org/miccai-2026/0584-Paper3584.html)
- Built a training package that allows for the continual retraining of a model as samples are approved by radiologists using LinGuinE, increasing the size of our training data and allowing other workflows to utilise these larger datasets.
- Performed survival modelling on clinical trial datasets. Experimentation focused on fusing modalities across baseline patient information, CT embeddings and H&E slide embeddings. Using dimensionality reduction and gated fusion techniques, C-Index scores were raised from 0.66 to 0.71.
- Implemented and fine-tuned a version of MedSAM2 to be utilised by the team as another segmentation model. Other models used include SwinUNETR and VISTA3D
- Collaborated with Microsoft Research to implement and evaluate BiomedParse v2 on internal clinical trial datasets for lung lesion segmentation. Fine-tuned the model's mask decoder and benchmarked performance against existing segmentation models; the fine-tuned model did not outperform the established baselines.
- Mentored rotational colleagues to improve their abilities with python and expose them to cutting edge machine learning research.

Kubrick Group

Machine Learning Engineer *2023 - Present*

- Completed a comprehensive 17-week training period, covering SQL, Docker, MLOps, Time Series Analysis, Spark, PyTorch, NLP, Deep Learning and command line environments.
- Developed a Random Forest model for predicting banking churn on an artificially generated dataset.
- Deployed solutions using FastAPI and Postman and containerised using Docker
- Collaborated in small teams using git, GitLab and MS Azure.

NextWave Consulting

Associate Consultant (Appian Developer) *2022 - 2023*

- Achieved Appian Associate Developer Certification
- Led the development of an internal onboarding tool. This included designing a coherent and efficient data model, before building it using MySQL
- Placed at a FTSE 100 insurer, building an internal tool to automate processes and data pipelines. Our progress was communicated to non-technical stakeholders, including senior figures, business analysts and end users.

EDUCATION

University of Bath

BSc (Hons) Physics with Year Long Work Placement *2018 - 2022*

- Awarded Upper Second Class Honours.
- Highlights include 80% in Mathematical Methods for Physics 2 (Linear Algebra, Statistical Modelling) and 77% in Quantum & Atomic Physics (Probability).
- Achieved a First for my Final Year Project, where I worked in a small team with the Met Office to build a neural network to predict water wave heights from a nearby weather radar using TensorFlow and Py-ART.

Other Achievements

- Awarded an AWS Scholarship to study the AI programming with Python Nanodegree via Udacity (2022)
  - Built deep learning models, such as a flower classifier using transfer learning from VGG16 in PyTorch
  - Selected for the Advanced Program (Top 25%), building models on AWS using PyTorch
- Deep Learning Specialisation and Browser-based TensorFlow models course, deeplearning.ai (2020)
- UKMT Senior Team and Individual (Silver) Maths Challenge (2018)